Deploying and Managing WSUS

(Windows Server Update Services)

1. What is a Patch?

A patch is a small update file. It fixes a problem in a program, or it makes the program better.

A patch can do these things:

- Fix a security problem (something a hacker could use to break in)
- Fix a bug (something that is not working correctly)
- Make the program run faster
- Add a small new feature
- Make the system more stable (less likely to crash)

Example: Microsoft finds a security problem in Windows. Microsoft creates a patch to fix it. When you install the patch, your computer becomes safer.

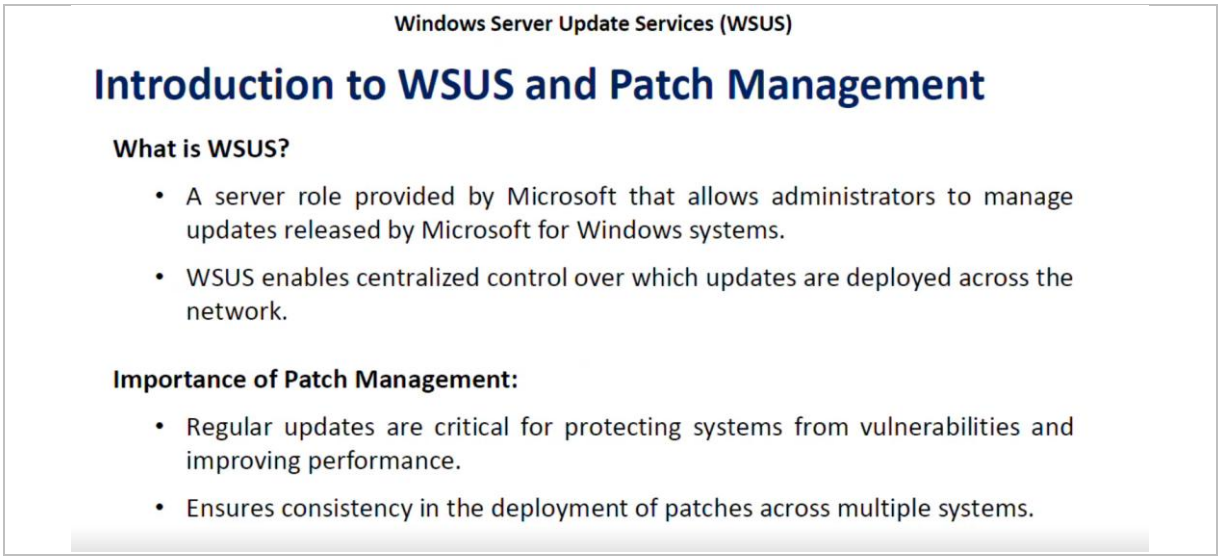


Figure 1: Introduction to WSUS and patch management

2. What is a WSUS Server?

WSUS stands for Windows Server Update Services. It is a free Microsoft tool. It helps a company manage Windows updates for many computers at once.

Normally, every computer would download updates from the internet by itself. With WSUS, only the WSUS server downloads the updates one time. Then it shares those updates with every

computer on the company network. This saves internet bandwidth and gives the IT admin more control.

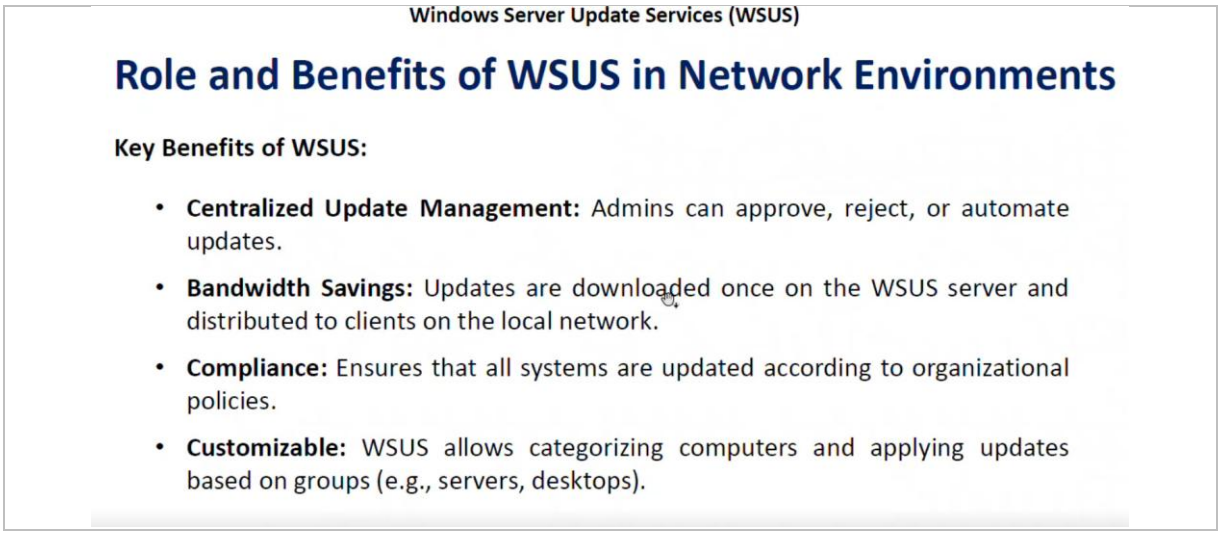


Figure 2: The role and benefits of WSUS in a network

The main advantage of WSUS is centralized patch management. This means one admin can control updates for the whole company from one place.

On a normal home PC:

- Windows downloads and installs updates by itself, with little control from the user.
- You cannot easily control which updates go to which computer.

In a company (corporate) network:

- The IT admin needs full control over which updates are installed, and when.
- WSUS gives the admin this control.

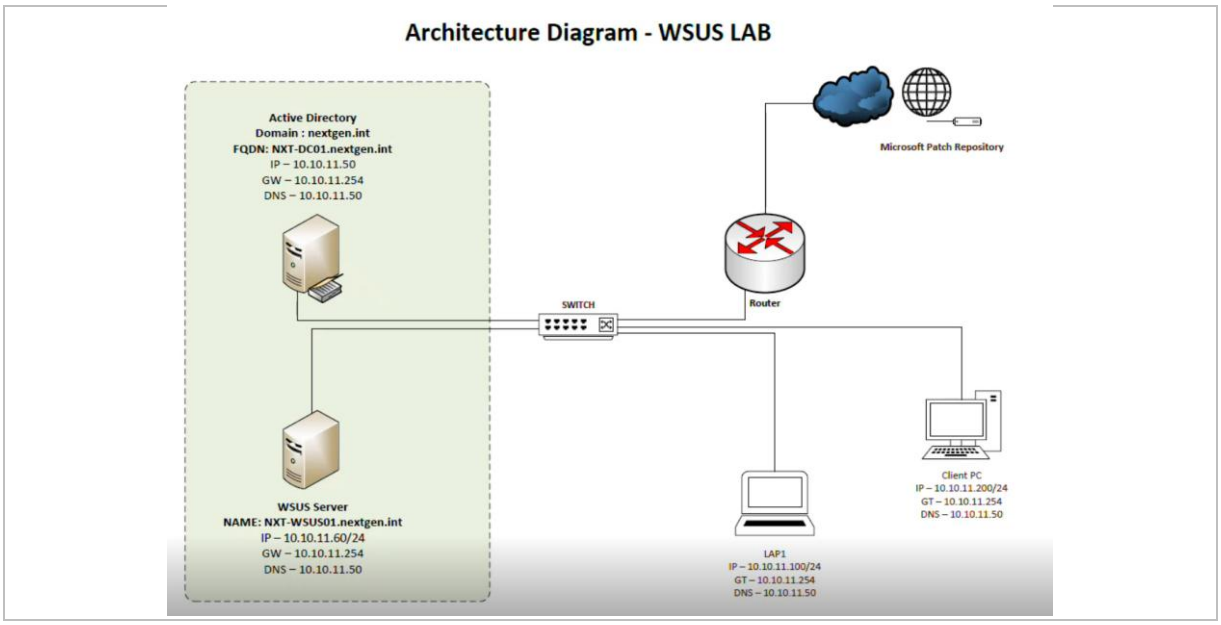


Figure 3: Example WSUS lab architecture

3. What is the Microsoft Patch Repository?

The Microsoft Patch Repository is Microsoft's own online storage. It is like a giant online library of every Windows update ever released.

Whenever Microsoft creates a new update, it is placed in this repository. From there, normal Windows computers and WSUS servers can download it.

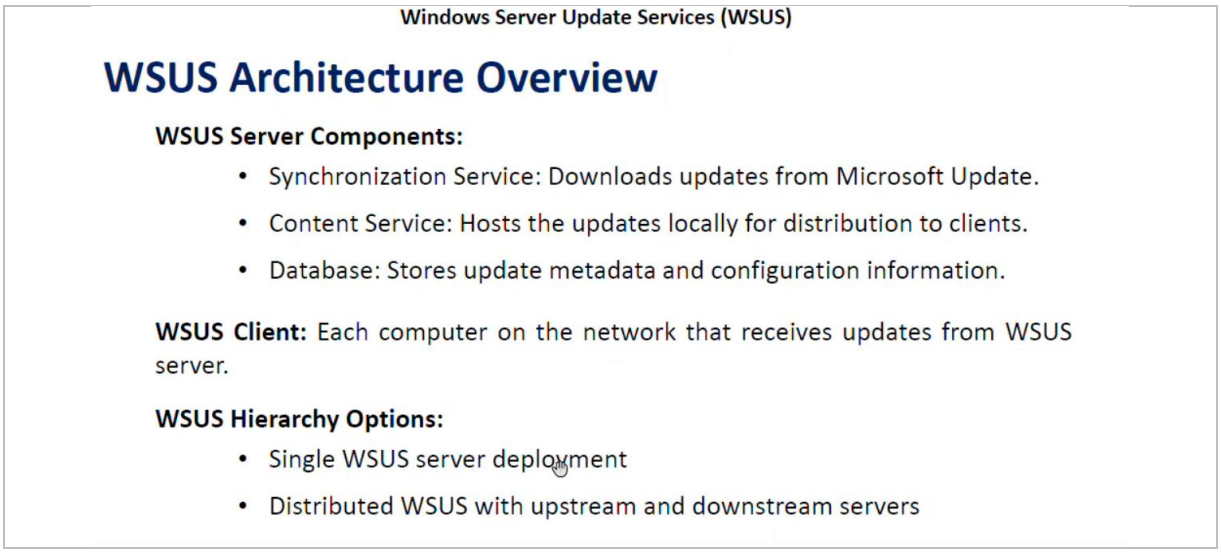


Figure 4: WSUS architecture overview

4. The Three Core Parts of WSUS

Part	What it does
Synchronization	The process where the WSUS server connects to Microsoft Update and checks: “Are there any new updates?”
Content	The actual update files themselves (file types like .cab or .msu). These are the files a client computer downloads and installs.
Database	Stores information about the updates, such as their names and status. It does not store the update files themselves — those are the “content” above.

Table 1: The three core parts of WSUS

NOTE *In almost every company, the WSUS server is joined to the Active Directory (AD) domain. This is a recommended best practice, but it is not a strict technical requirement.*

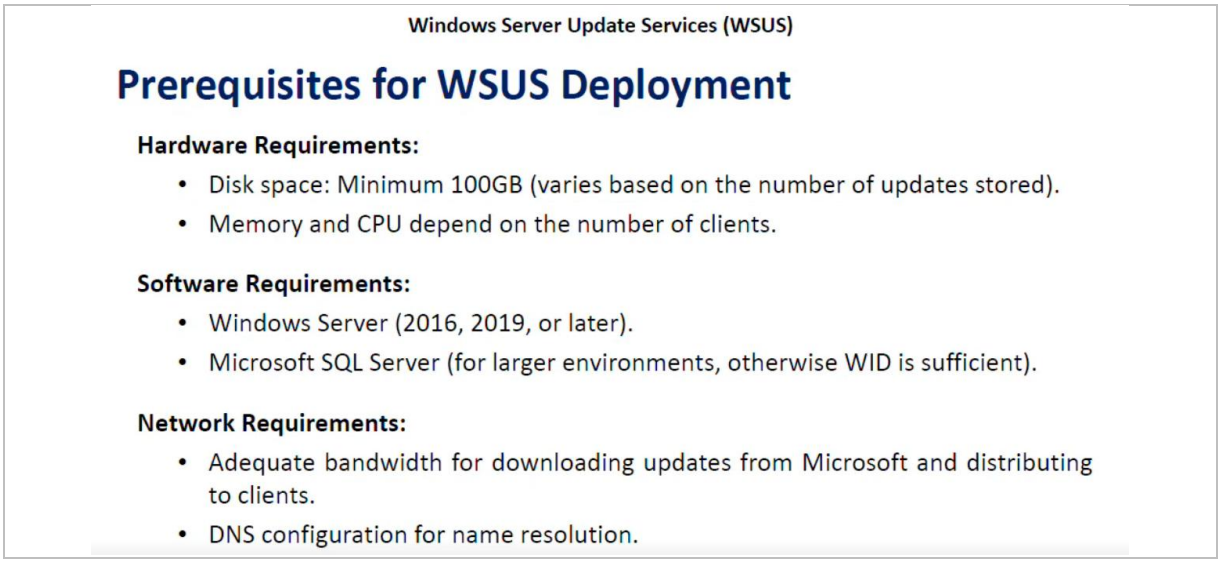


Figure 5: Prerequisites for a WSUS deployment

5. Installing the WSUS Server

Follow these steps to install the WSUS role on Windows Server.

- Open Server Manager, click “Add Roles and Features”.
- On the server roles page, select “Windows Server Update Services”.

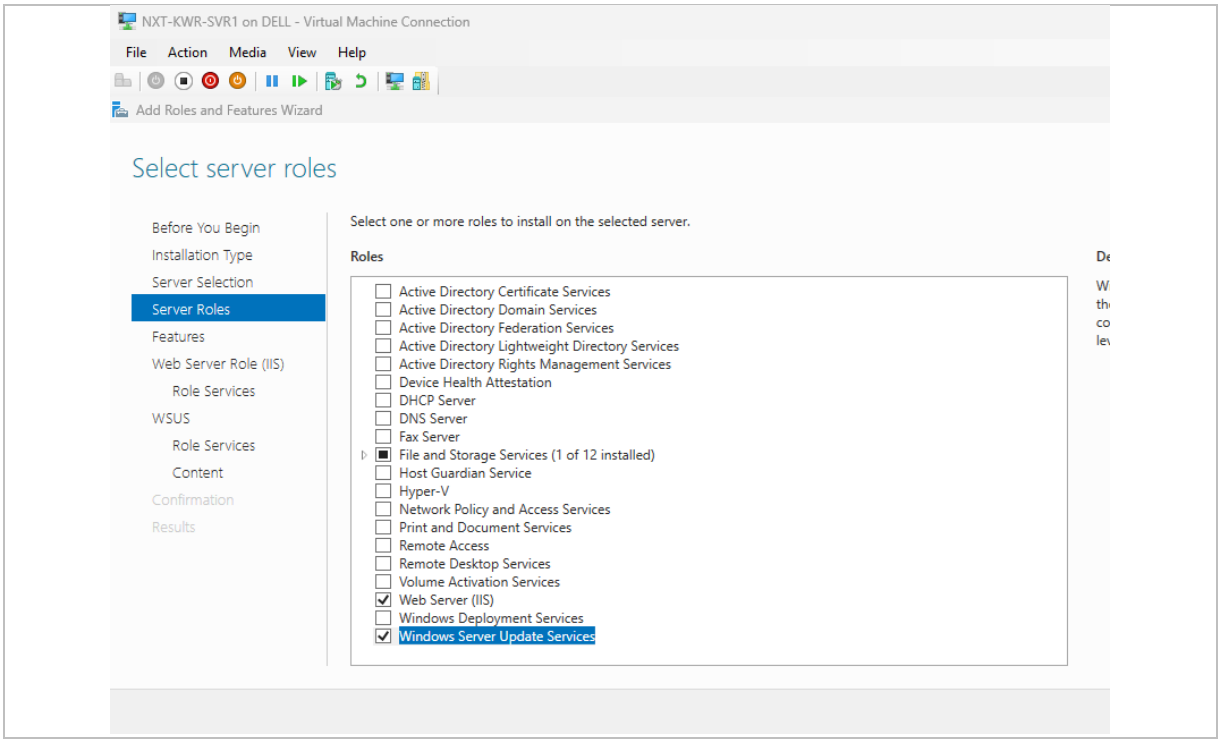


Figure 6: Selecting the WSUS server role in the Add Roles and Features Wizard

Click “Add Features” when it asks, then click Next.

> **NOTE** *You may notice that Web Server (IIS) gets selected automatically. This is normal. WSUS needs IIS because IIS is the web server that publishes (hosts) the update files to client computers.*

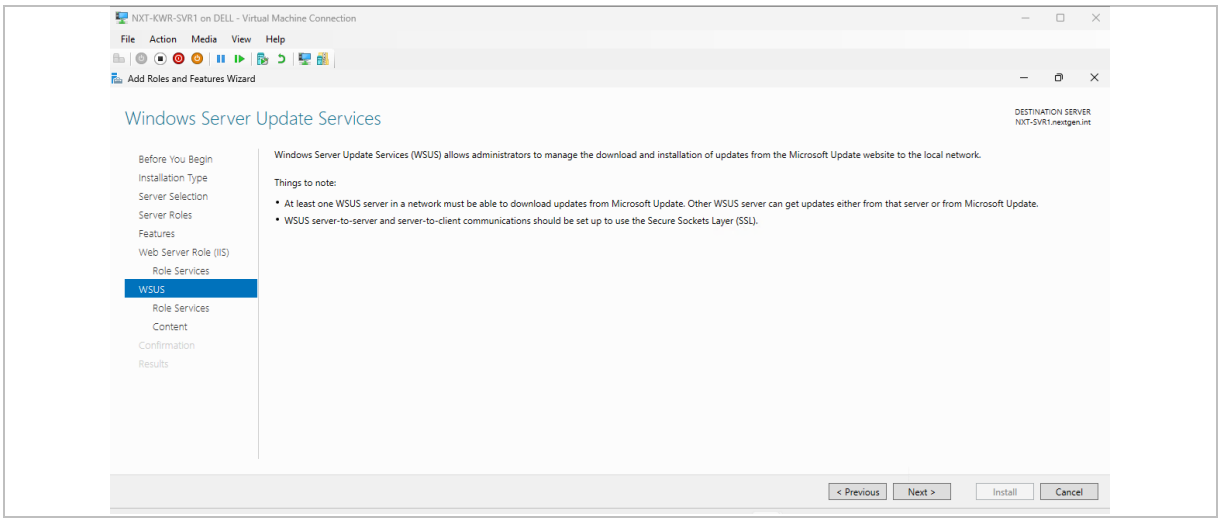


Figure 7: WSUS role description, showing the IIS requirement

6. Choosing How WSUS Stores Its Data

WSUS needs a database to store update information and a list of client computers. On the “Role Services” screen you will see three options. Here is what each one means, in simple words:

Option	What it means
WID Connectivity (usually checked)	WID stands for Windows Internal Database. It is a small, free database built into Windows. It needs no license and no extra setup — it just works automatically. This is the best choice for small or medium companies.
WSUS Services (must be checked)	This is the core engine of WSUS. It installs the actual management tools and the interface you use to control updates. This option is required — without it, WSUS is not really installed.
SQL Server Connectivity (usually unchecked)	This lets WSUS connect to a full, separate Microsoft SQL Server instead of WID. Large companies with tens of thousands of computers use this option, because WID can

Option	What it means
	become too slow at that scale. A full SQL Server is faster and supports advanced setups, like clustering multiple WSUS servers together.

Table 2: WSUS role service options

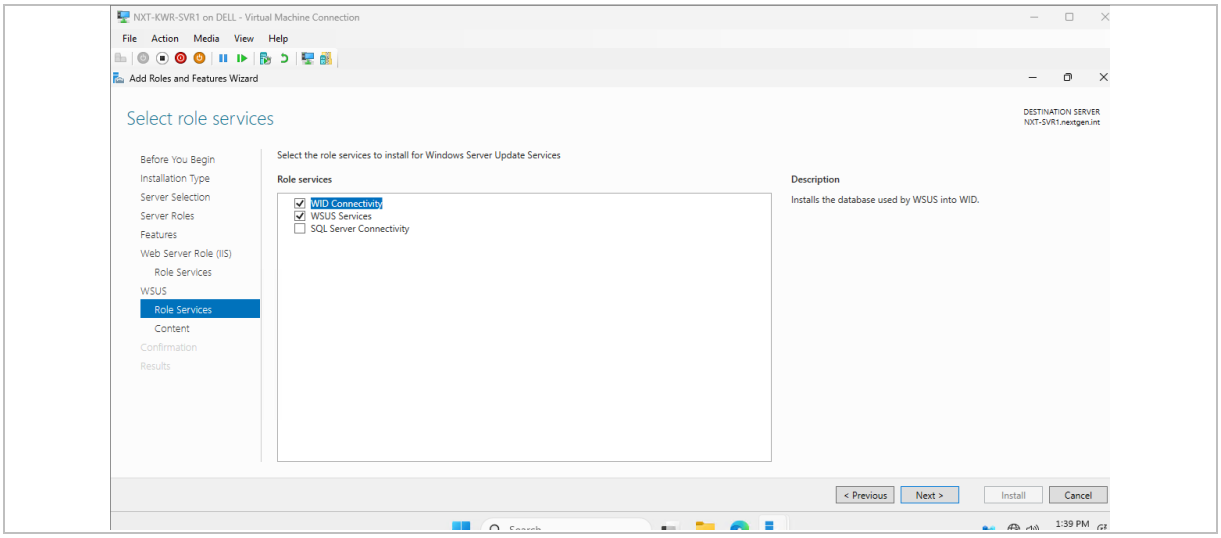


Figure 8: Choosing WSUS role services

NOTE *When you choose a folder path to store update files, do not choose the C: drive (the drive where the operating system is installed). Choose a different drive, so update files do not fill up your OS drive.*

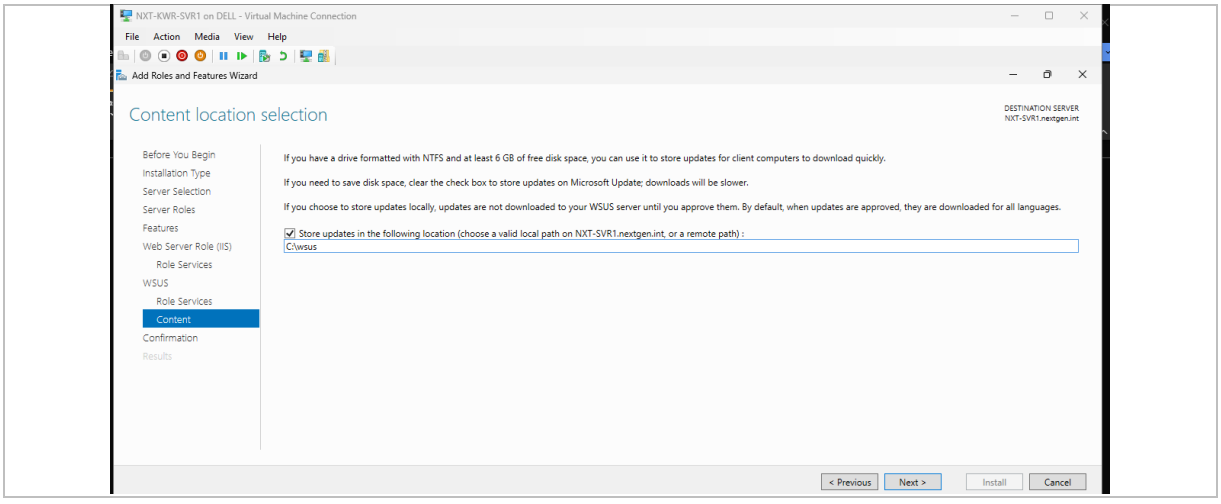


Figure 9: Choosing the content storage location

7. Configuring WSUS After Installation

After WSUS is installed, we need to configure it manually. First, open the WSUS console.

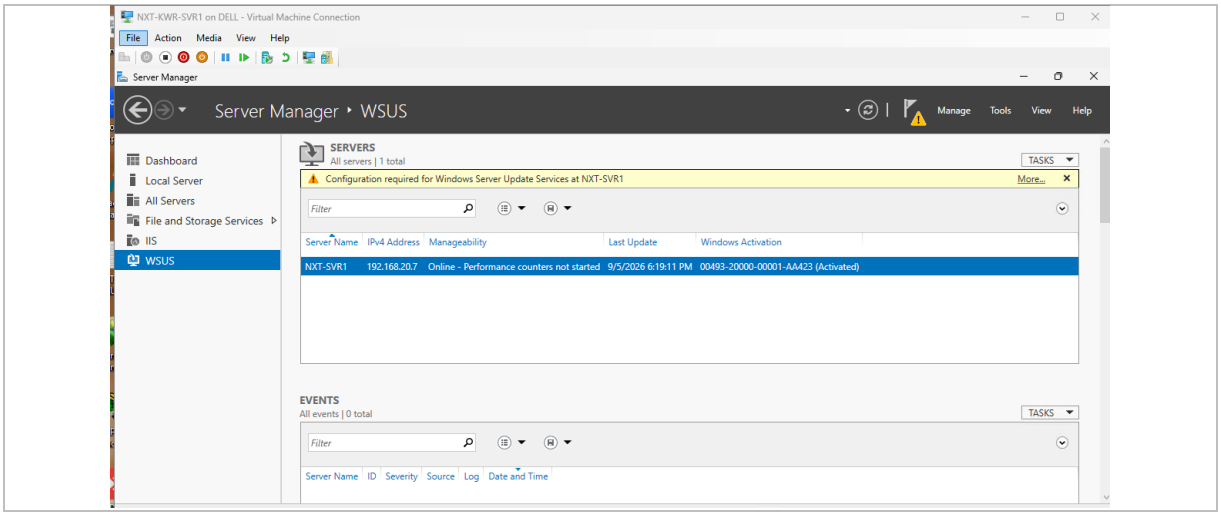


Figure 10: Server Manager showing the WSUS role added

Right-click the server name, then click “Windows Server Update Services” to open the configuration wizard.

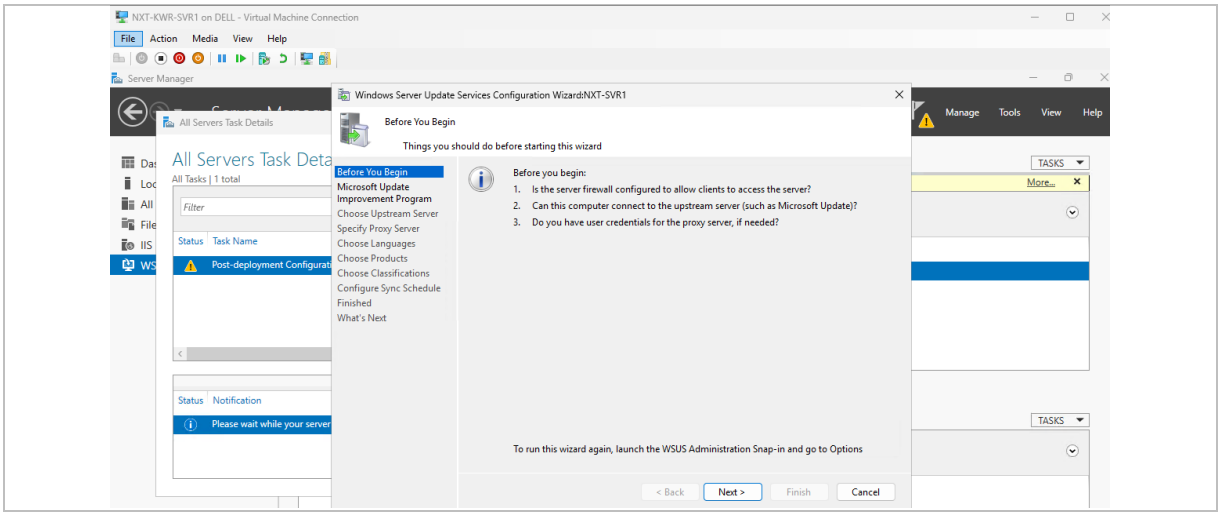


Figure 11: WSUS Configuration Wizard — Before You Begin

Before you start: three important questions

1. “Is the server firewall configured to allow clients to access the server?”

Simple meaning: A firewall is like a security guard for your server — it blocks traffic it does not recognize. WSUS works like a website (because it runs on IIS), so client computers must be able to reach it. This usually means opening Port 8530 (normal HTTP) or Port 8531 (secure HTTPS) on the server's firewall. If these ports are blocked, client computers cannot talk to WSUS, and updates will fail.

2. “Can this computer connect to the upstream server (such as Microsoft Update)?”

Simple meaning: Your WSUS server needs internet access to reach Microsoft's servers, so it can download update information before sharing it with your local computers. If there is no internet access, or your company firewall blocks it, WSUS cannot see or download any new patches.

3. “Do you have user credentials for the proxy server, if needed?”

Simple meaning: Some companies do not allow computers to connect to the internet directly. Instead, computers must go through a proxy server (a kind of middleman). If your company uses a proxy, WSUS will need the proxy's IP address, plus a username and password, so it can pass through and reach Microsoft.

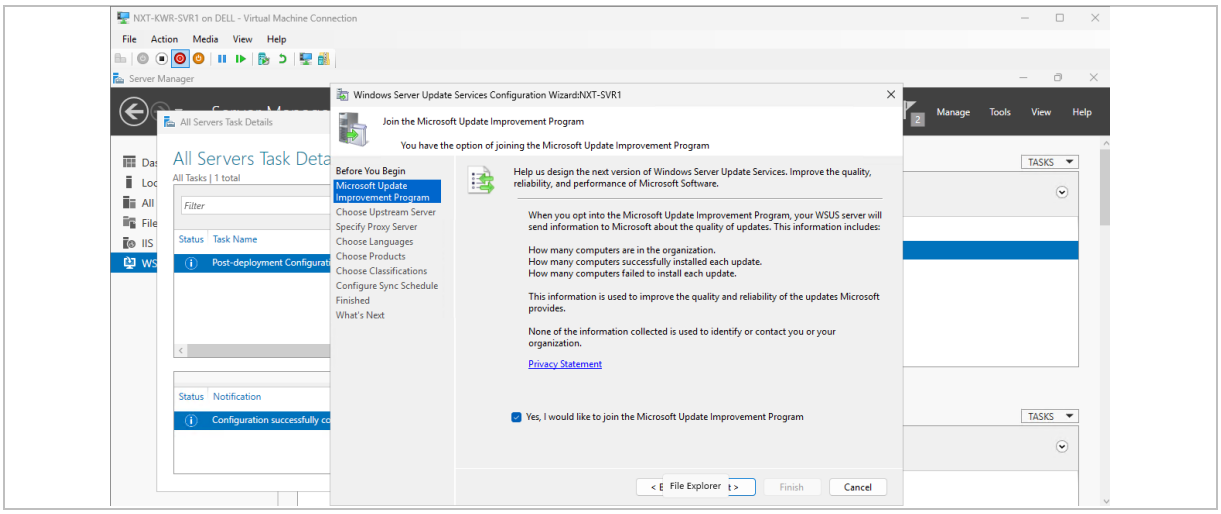


Figure 12: Microsoft Update Improvement Program screen

Microsoft Update Improvement Program

This screen asks if you want to join Microsoft's improvement program. You can click Next without ticking the box. If you want to join, that is fine too — it is optional.

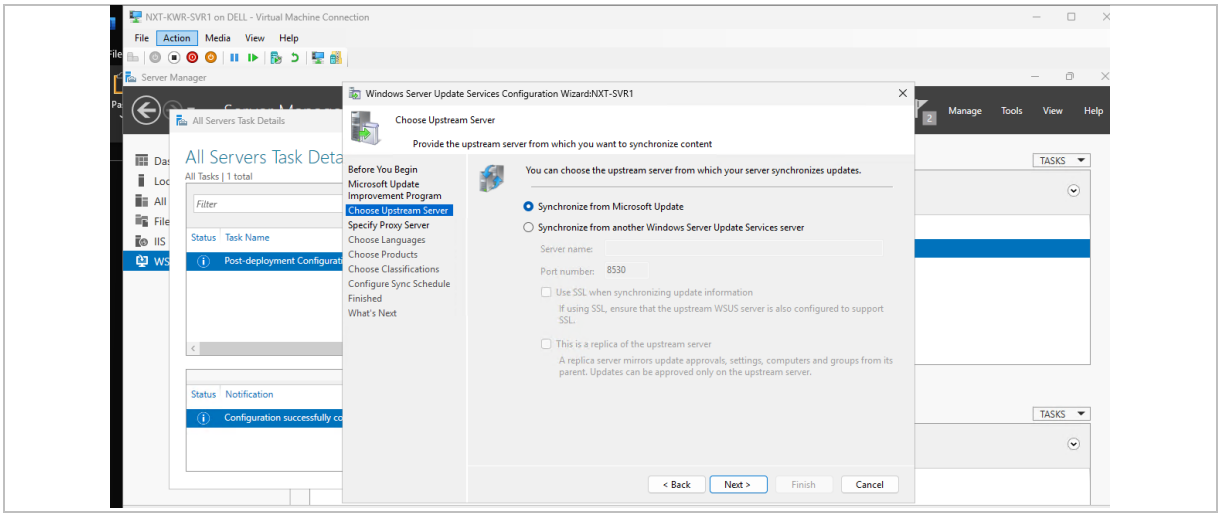


Figure 13: Choose Upstream Server

8. Choosing Where WSUS Gets Its Updates

Option 1: Synchronize from Microsoft Update

Your WSUS server connects directly to Microsoft's servers over the internet to download update data. Use this option if this is the only WSUS server in your company, or if it is the main “upstream” server at head office.

Option 2: Synchronize from another WSUS server

Instead of using the internet, this WSUS server connects to another WSUS server already running on your company network. This is useful for companies with branch offices. For example, if head office is in New York and you are setting up a new branch server in London, the London server can download updates directly from the New York server. This saves internet bandwidth for the London office.

When you choose this second option, three extra settings appear:

- Server name and port: Enter the network name of the main (master) WSUS server, plus the port it uses (usually 8530 or 8531).
- Use SSL when synchronizing update information: This encrypts the traffic between the two WSUS servers, so no one on the local network can spy on it.
- This is a replica of the upstream server: If left unchecked (Autonomous mode), this server downloads the files but you still approve or decline updates on it yourself. If checked (Replica mode), this server becomes a full mirror of the main server — whatever you approve or configure on the main server is copied here automatically.

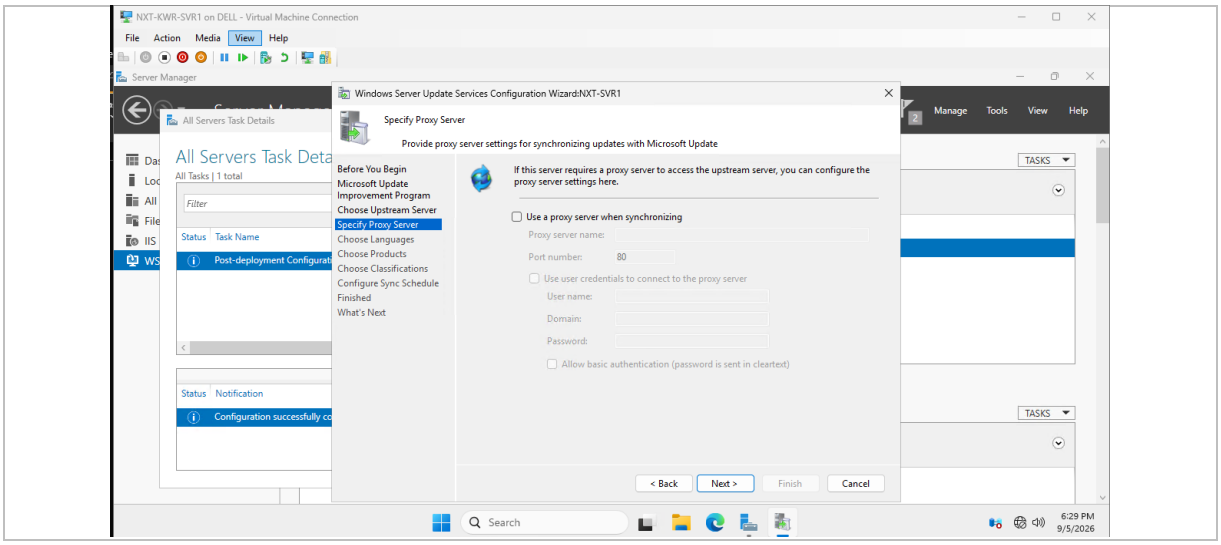


Figure 14: Specify Proxy Server

Proxy Server Settings

If your network uses a proxy server, enter the username, domain, and password here. If you do not use a proxy, just click Next.

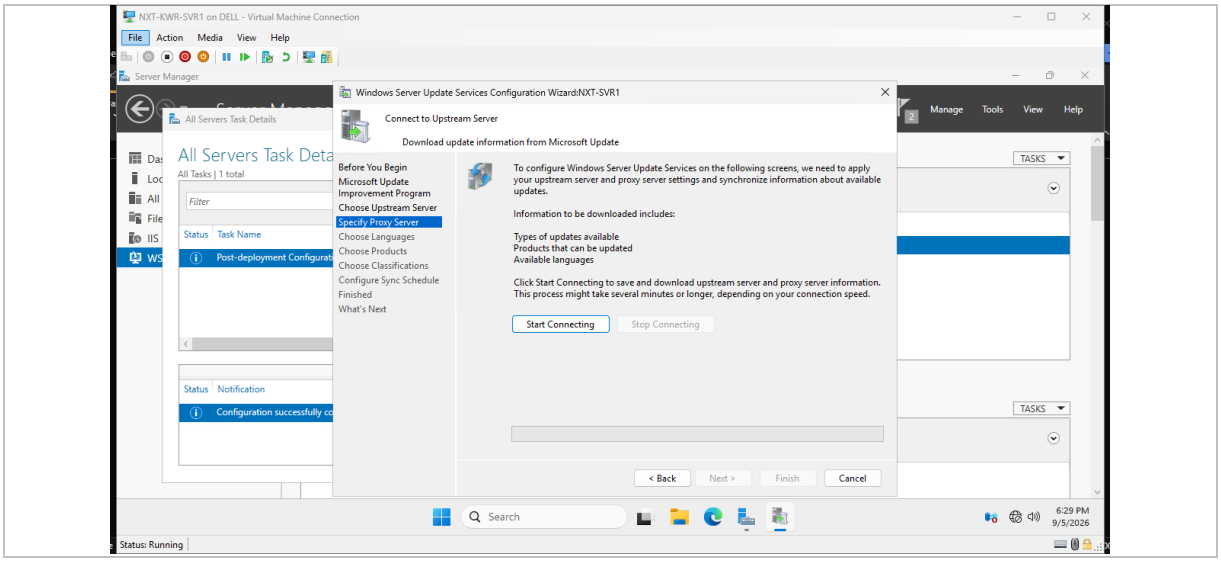


Figure 15: Connect to Upstream Server — initial synchronization

Connecting to Microsoft (Initial Synchronization)

When you click “Start Connecting”, the WSUS server contacts Microsoft's servers for the very first time. This is called the initial synchronization. WSUS downloads three things during this step:

- Types of updates available — the update categories, such as critical updates, security patches, or feature packs.
- Products that can be updated — a full list of Microsoft products, such as Windows 10, Windows 11, SQL Server, and Microsoft Edge.
- Available languages — the list of languages Microsoft supports for these updates.

If your firewall is blocking the connection, or your proxy settings are wrong, this step will fail. Think of it as a gatekeeper: it proves your internet connection is working correctly before you continue.

> **NOTE** *If synchronization fails at this point, WSUS setup cannot continue. Recap of the three main setup stages — Step 1: Role installation. Step 2: Post-installation tasks. Step 3: Configuration Wizard (this is where synchronization happens).*

Choosing a Language

After connecting, choose the language you want updates for. In this guide, we choose English.

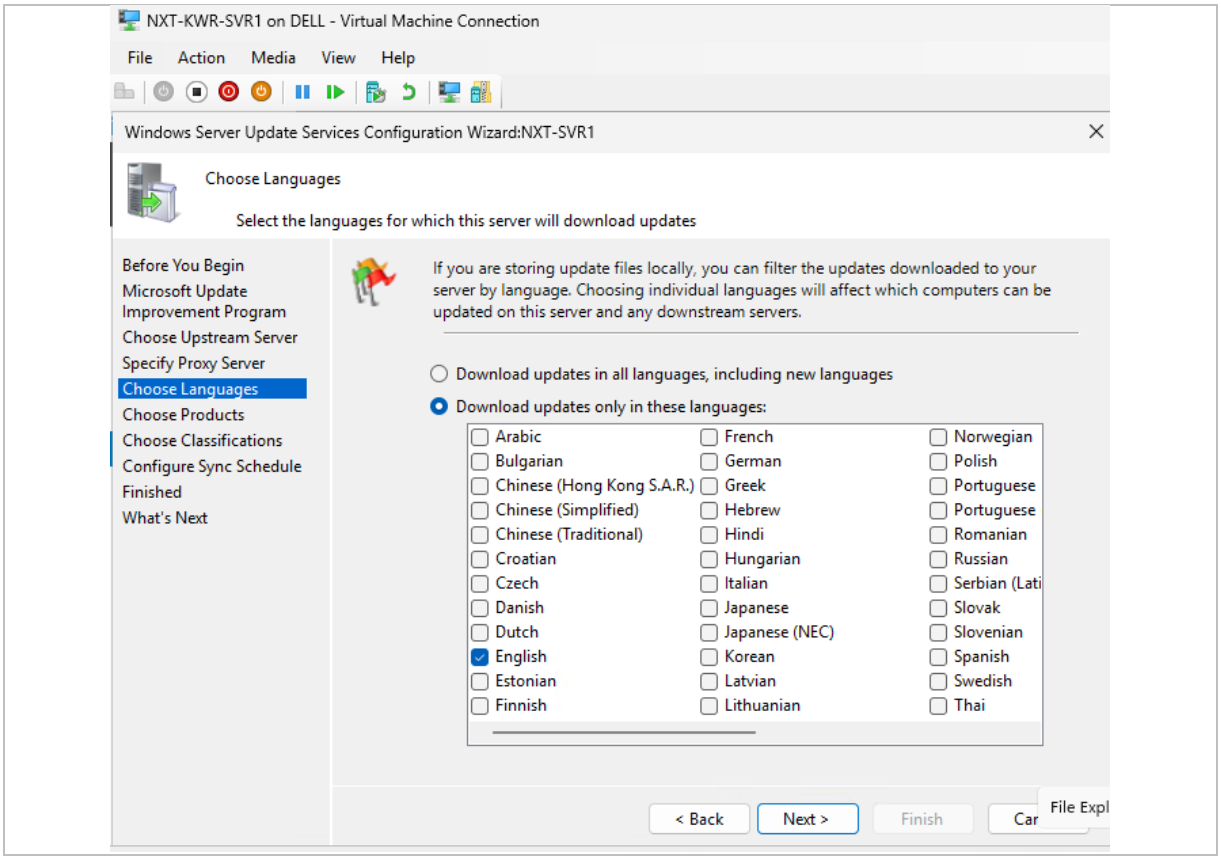


Figure 16: Choose Languages

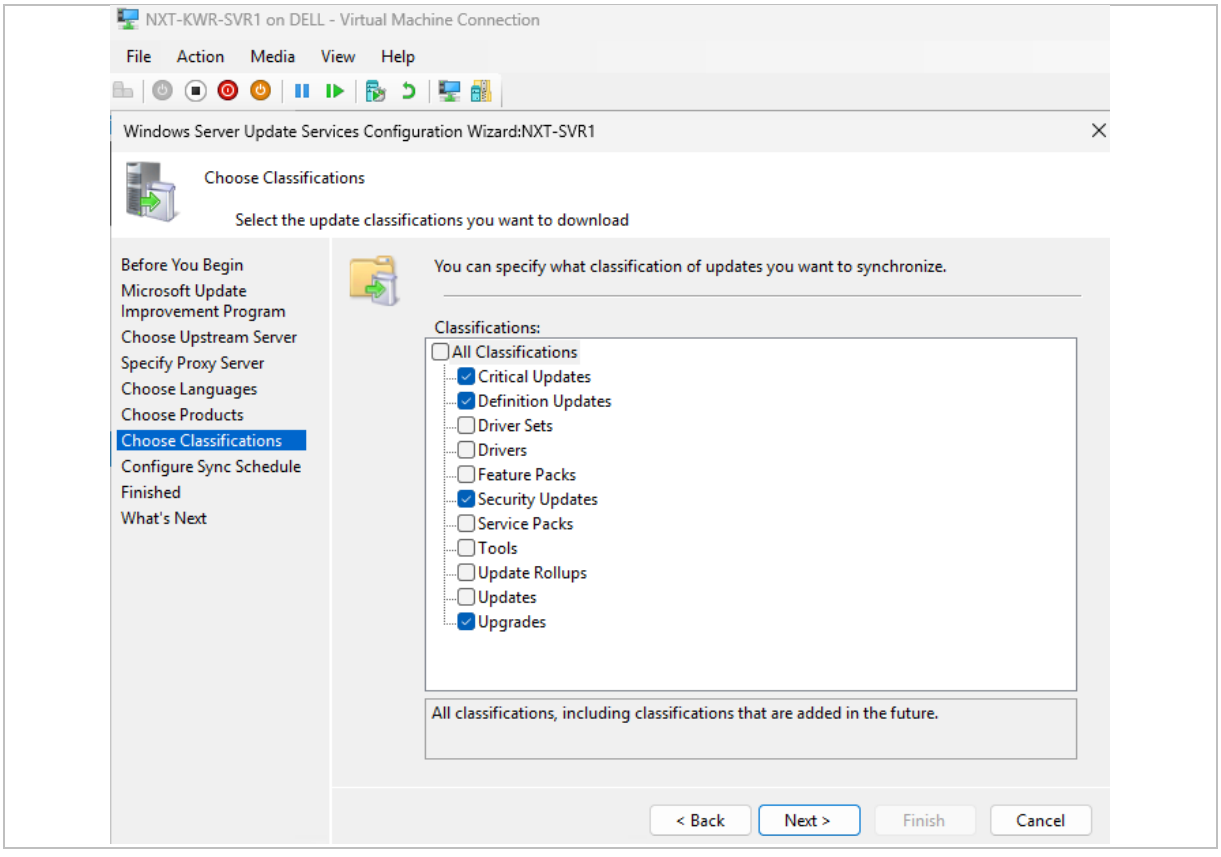


Figure 17: Choose Classifications

Choosing Products

Here you see a list of every Microsoft product. Select only the products your company actually uses (for example, Windows Server 2019 or Windows 11), so WSUS does not waste space downloading updates you do not need.

Choosing Classifications

Every patch belongs to a classification (category). In a corporate environment, it is common to select at least “Critical Updates” and “Security Updates”.

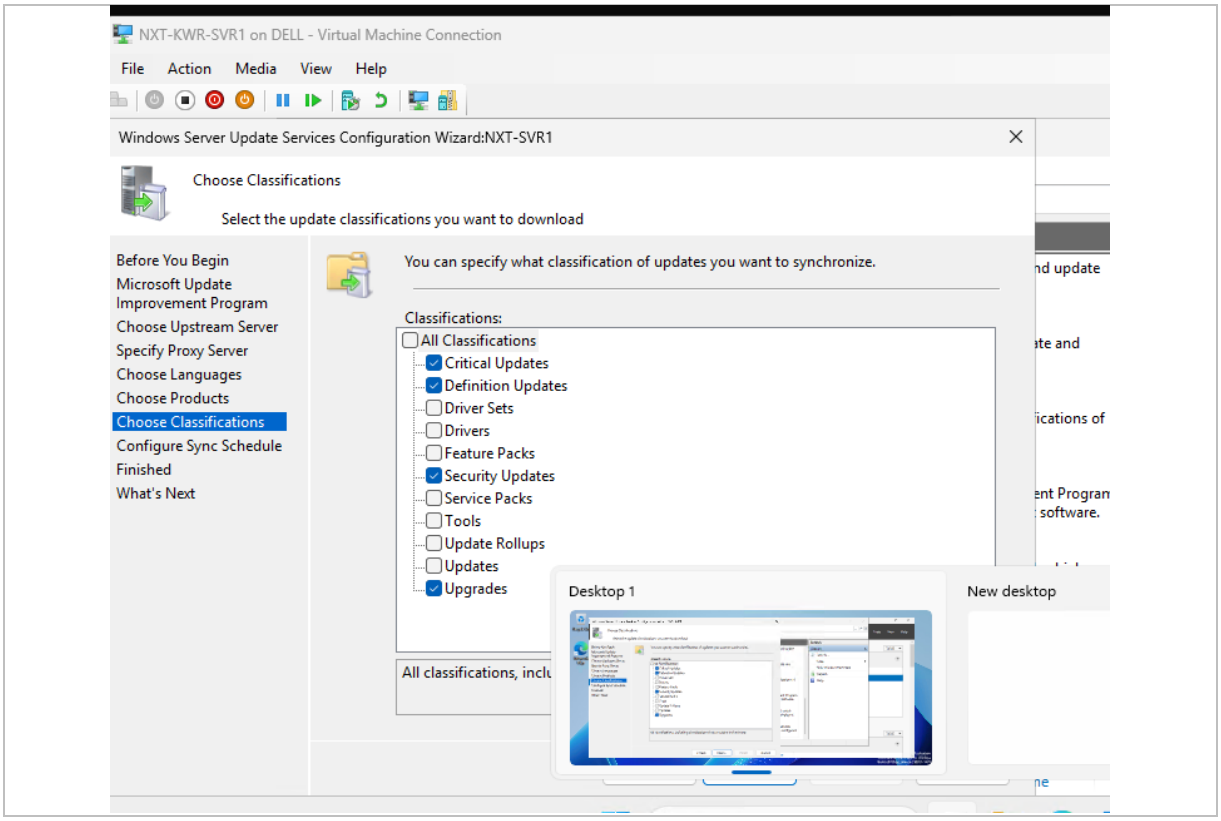


Figure 18: Confirming update classifications

Setting the Synchronization Schedule

Here you choose whether WSUS should synchronize manually or automatically, and how many times per day. Many companies leave the default schedule, which usually happens on Tuesdays (this lines up with Microsoft's regular “Patch Tuesday” release day).

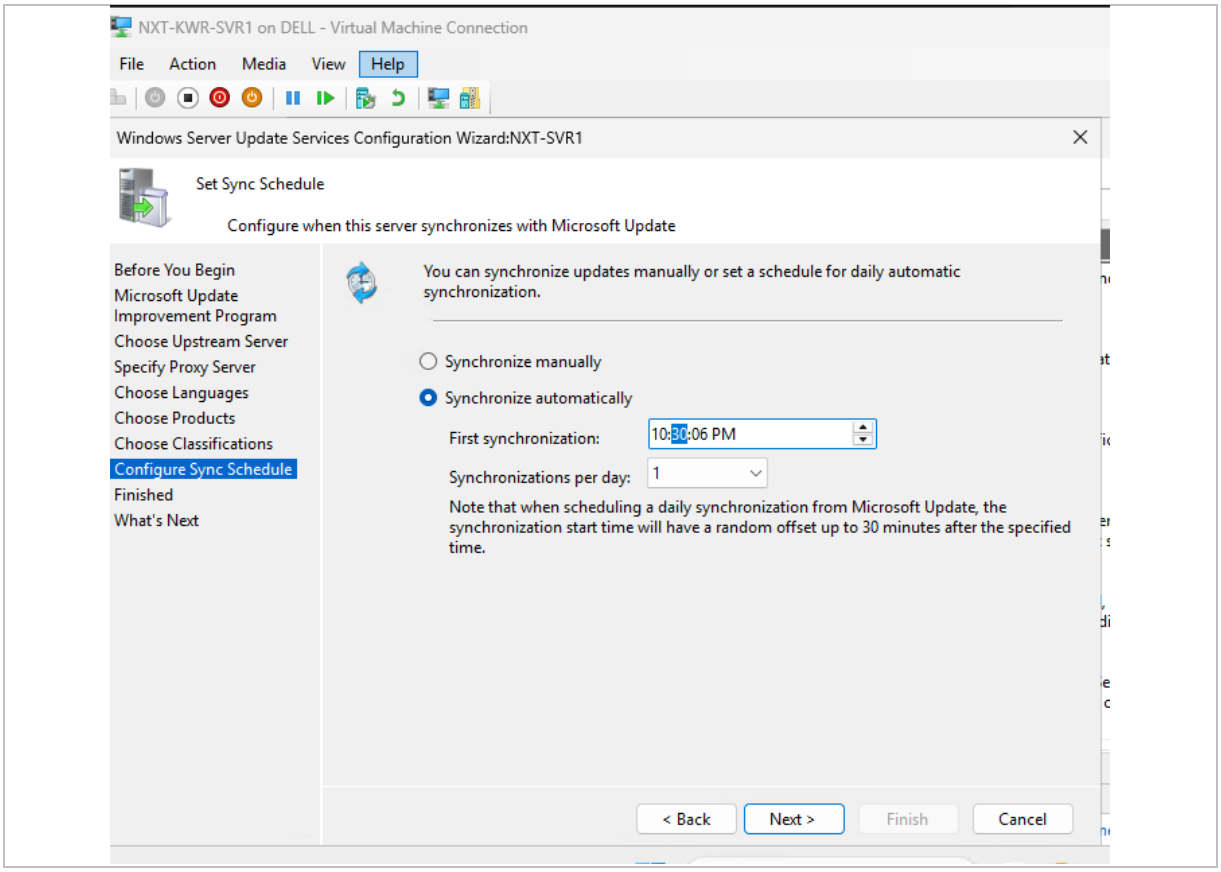


Figure 19: Set Synchronization Schedule

Finishing the Wizard

At the end of the wizard, you are asked if you want to begin the initial synchronization now. Tick the box, then click Next.

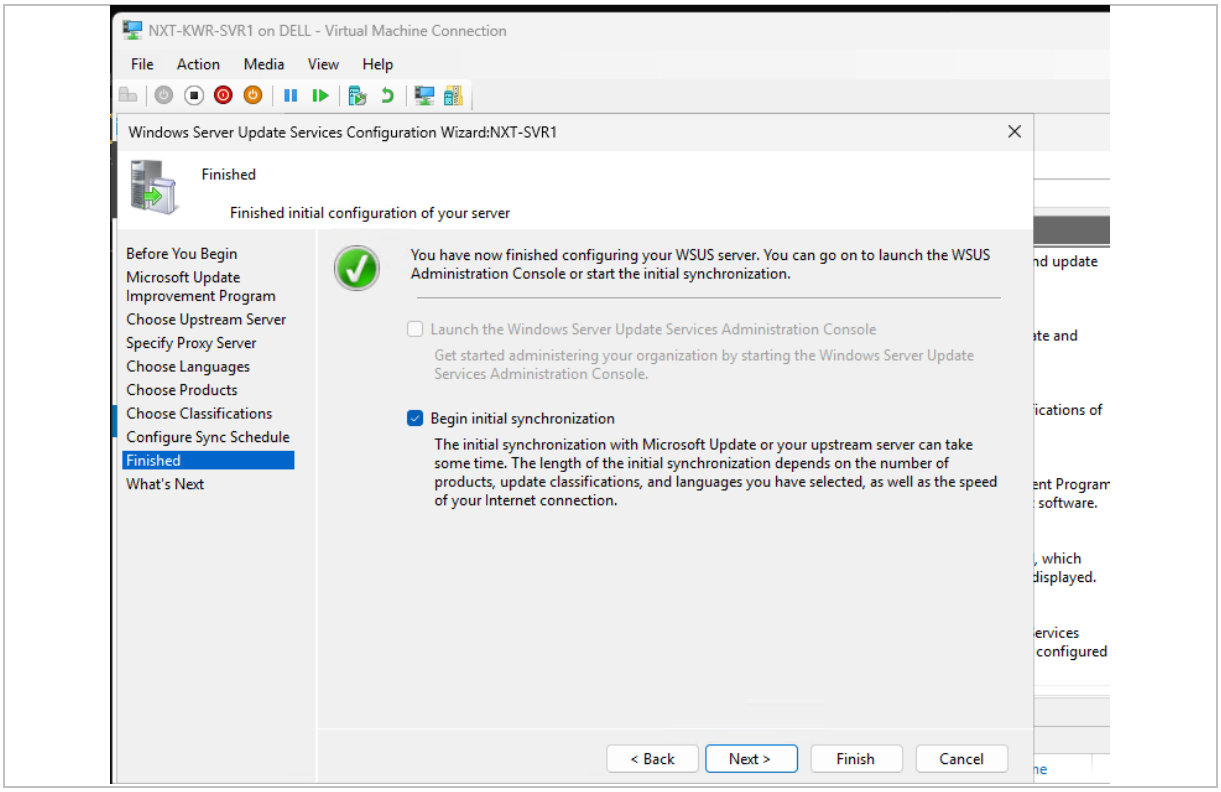


Figure 20: Finished — completing the configuration wizard

9. Reviewing Synchronization Results

The first synchronization only downloads a list of available patches — it does not download the actual patch files yet. At this point, the "To Do" section will show no updates yet, because nothing has been approved.

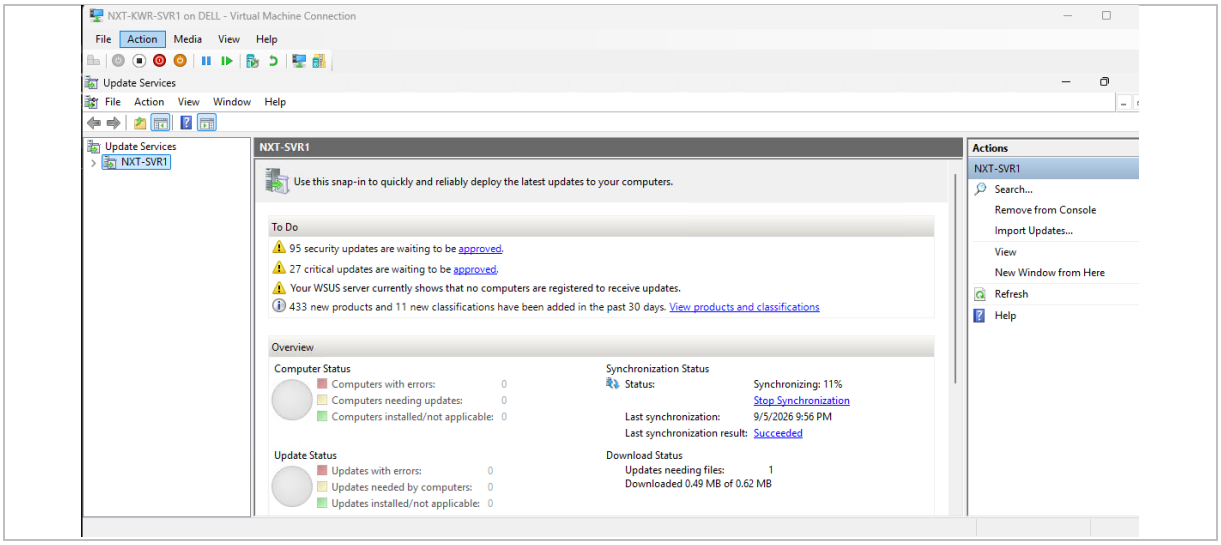


Figure 21: WSUS console — To Do list and synchronization status

After synchronization finishes, check the To Do section again.

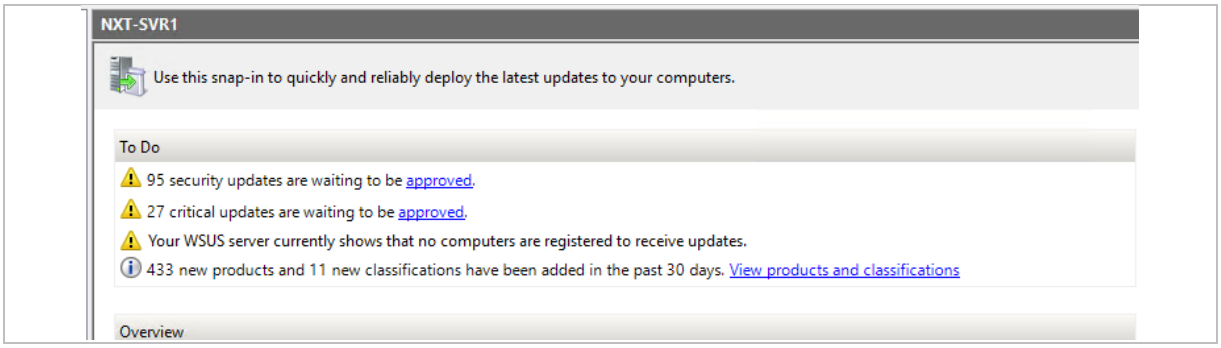


Figure 22: All Updates view after synchronization

Now you should see new updates listed.

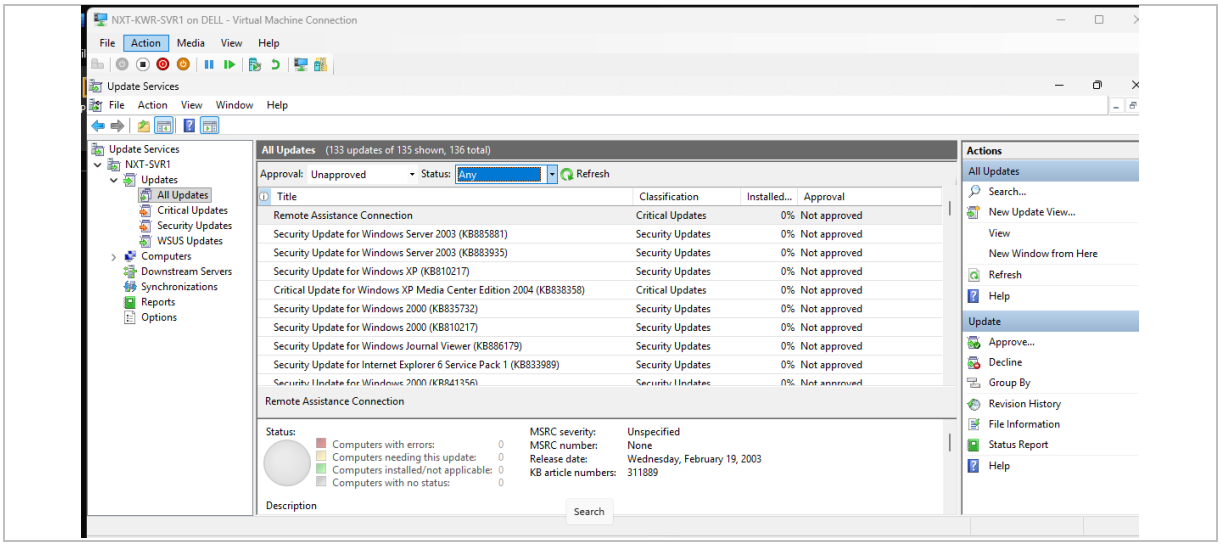


Figure 23: Updates ready to be approved

In this example, there are 34 security updates showing in the To Do section. To install the ones you need, right-click an update and click Approve. The update will then download and install on the server.

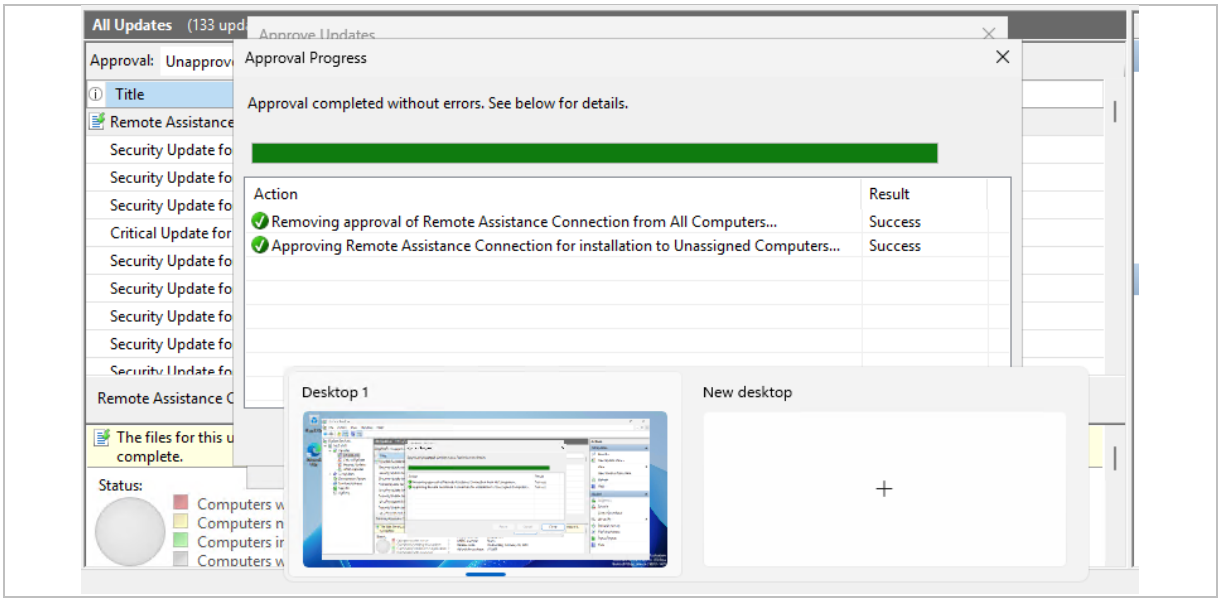


Figure 24: Approval progress after approving updates

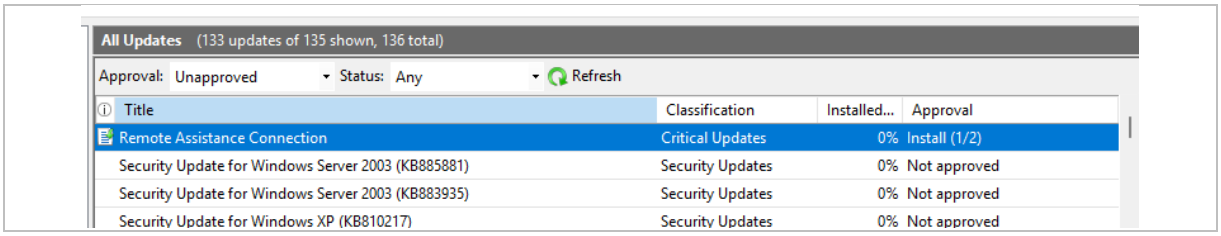


Figure 25: Update status changing to "Install" after approval

10. Managing Clients with Group Policy

Once WSUS is set up, we need to tell client computers to use it. The easiest way is with Group Policy in Active Directory.

Steps:

- In Active Directory, create an Organizational Unit (OU) — for example, one called "Servers".
- Inside it, create another OU for the specific systems, for example "Win 2025".
- Move the relevant computer accounts into this OU.
- Create and link a Group Policy Object (GPO) to this OU.

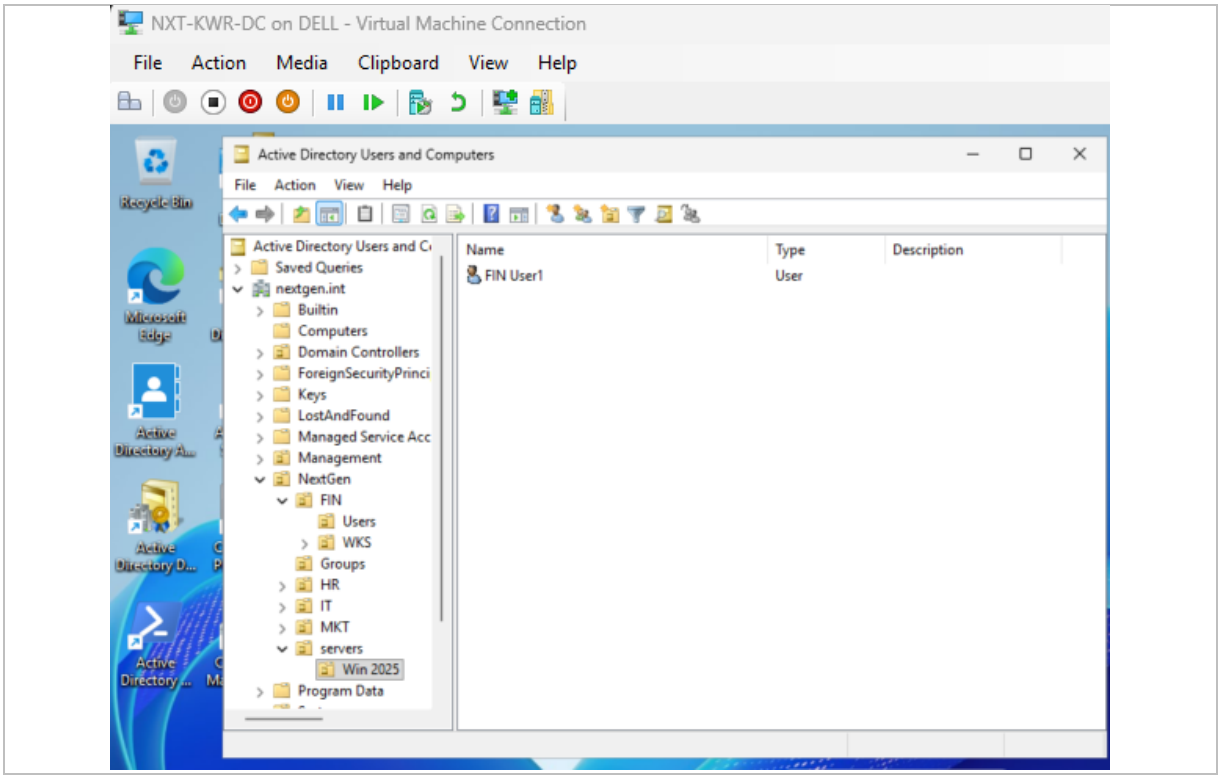


Figure 26: Active Directory Users and Computers — OU structure

Create a new Group Policy Object.

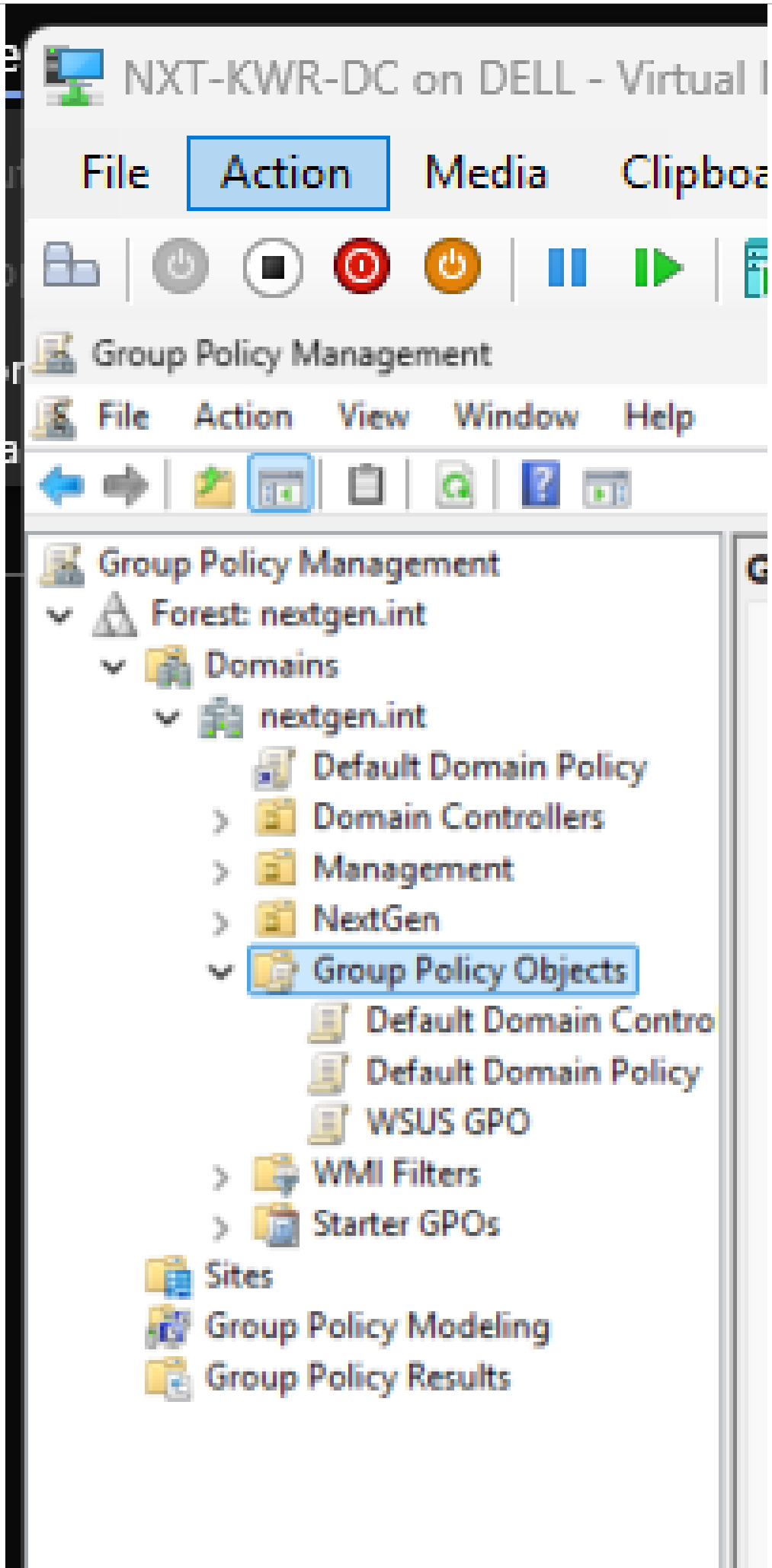

NXT-KWR-DC on DELL - Virtual
File
Action
Media
Clipboa
Group Policy Management
File
Action
View
Window
Help
Group Policy Management
Forest: nextgen.int
Domains
nextgen.int
Default Domain Policy
Domain Controllers
Management
NextGen
Group Policy Objects
Default Domain Contro
Default Domain Policy
WSUS GPO
WMI Filters
Starter GPOs
Sites
Group Policy Modeling
Group Policy Results

Figure 27: Group Policy Management — WSUS GPO linked in the domain

Right-click the GPO and click Edit. Then go to: Computer Configuration → Policies → Administrative Templates → Windows Components → Windows Update.

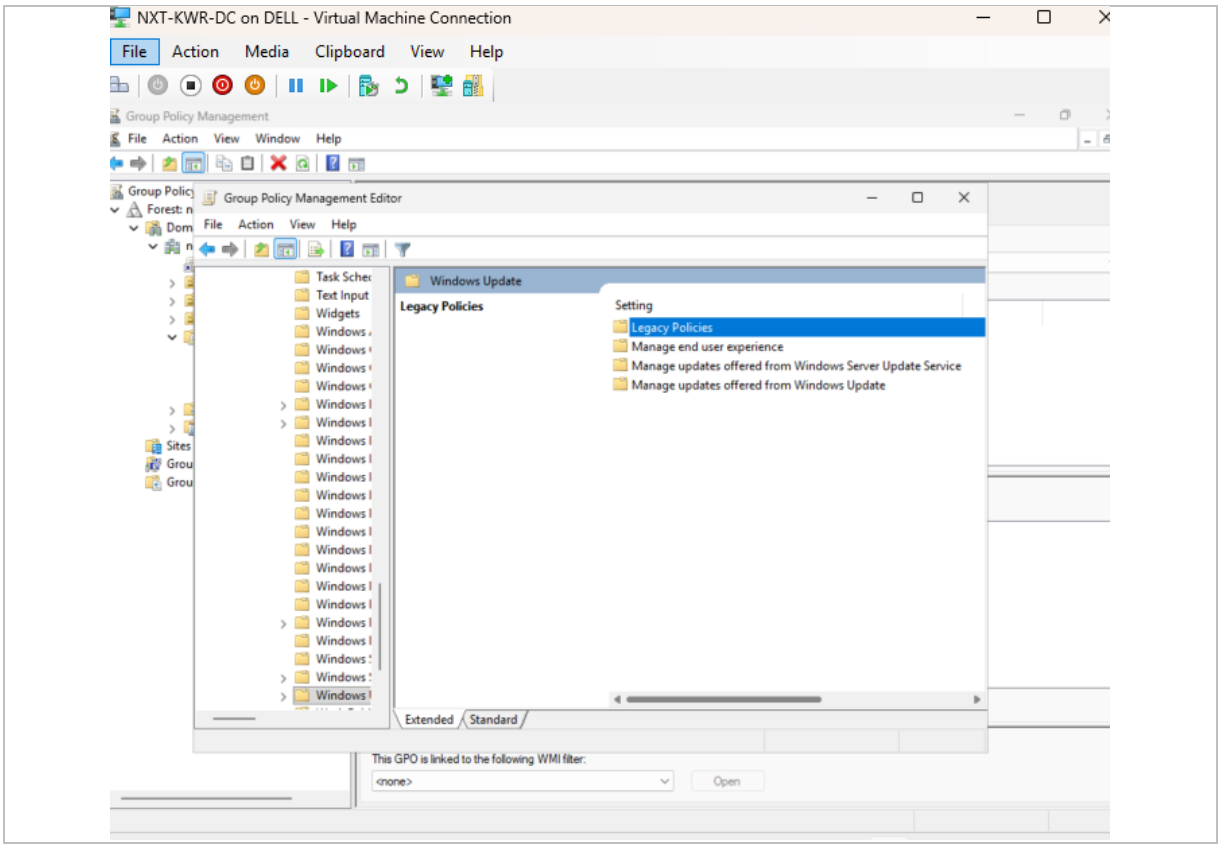


Figure 28: Group Policy Management Editor — Windows Update policies

Here you will find all the settings you can use to control how client computers get their updates.

Telling Clients to Use Your WSUS Server

Enable the policy called “Specify intranet Microsoft update service location” (in newer Windows versions this may be called “Manage updates offered from Windows Update”). This tells the client computers to get their updates from your WSUS server, instead of from the internet.

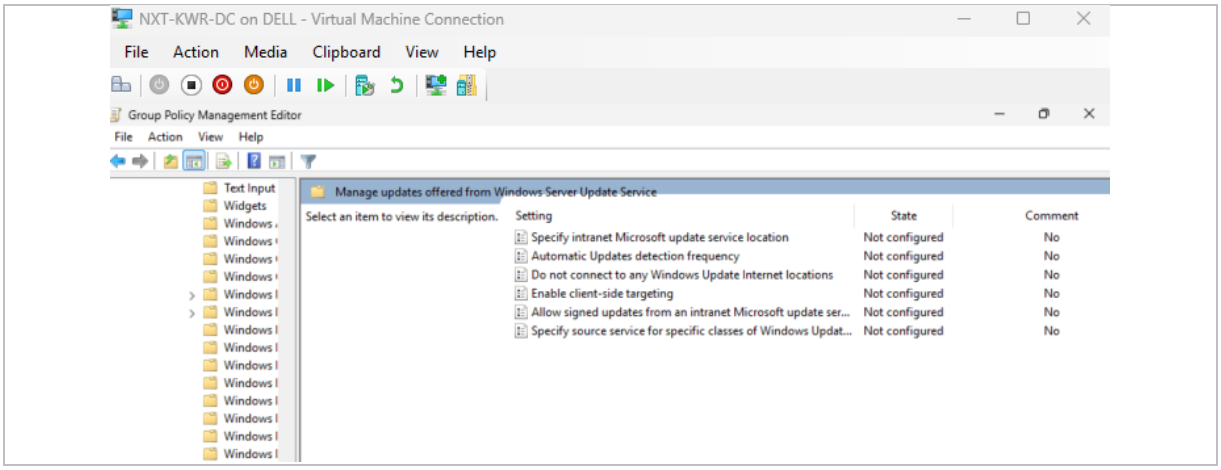


Figure 29: "Manage updates offered from Windows Server Update Service" settings

Click "Manage updates offered from Windows" and choose "Specify intranet Microsoft update service location".

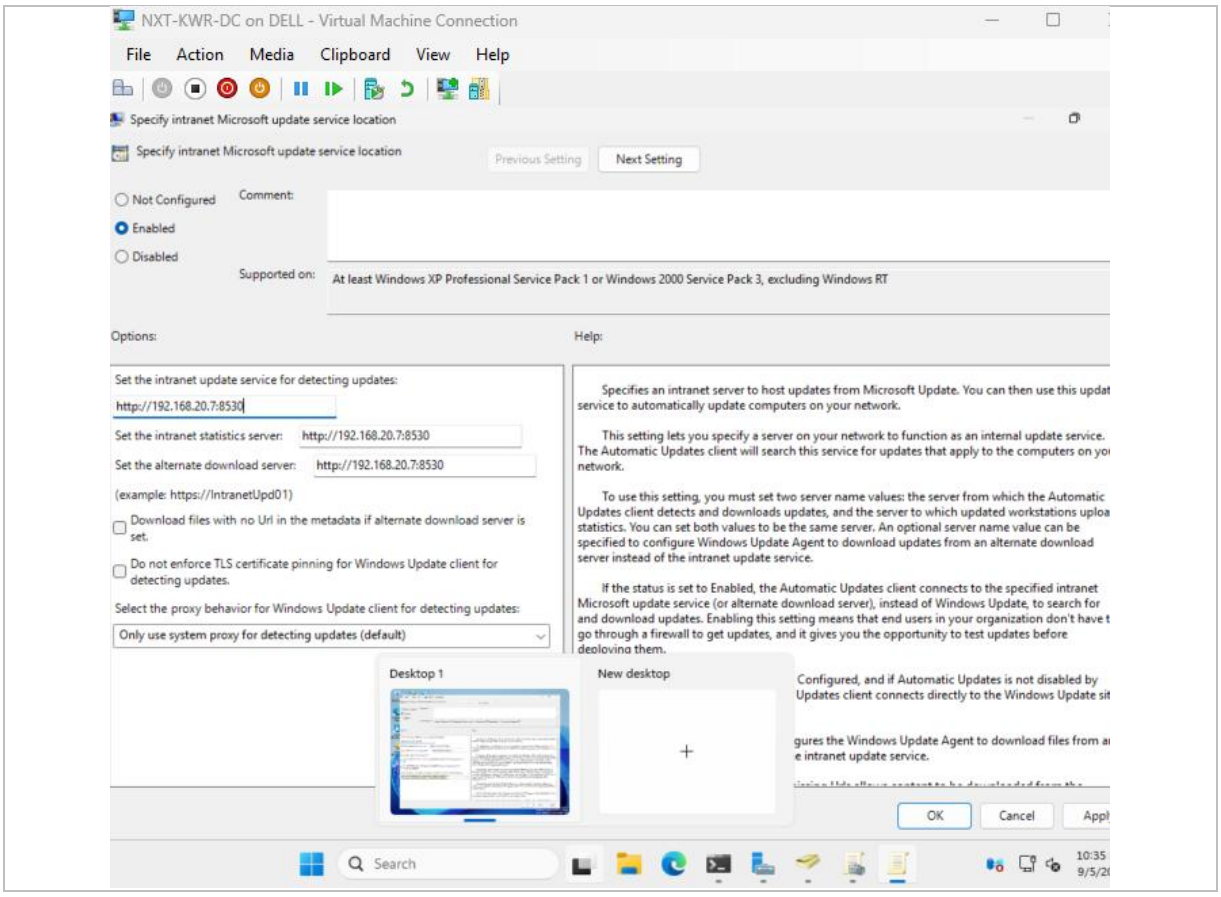


Figure 30: Specifying the intranet Microsoft update service location

Enter the URL of your WSUS server, for example http://192.168.20.7, and its port number, for example 8530.

Client-Side Targeting Policy

Client-side targeting is a WSUS feature. It automatically places a client computer into a specific computer group, using Group Policy — instead of the admin manually moving each computer into a group.

How it works: the admin creates a computer group on the WSUS server, and enters that same group name in the "Enable client-side targeting" policy setting. When a client computer refreshes its Group Policy and contacts WSUS, it is automatically added to the matching group. This saves the admin a lot of manual work.

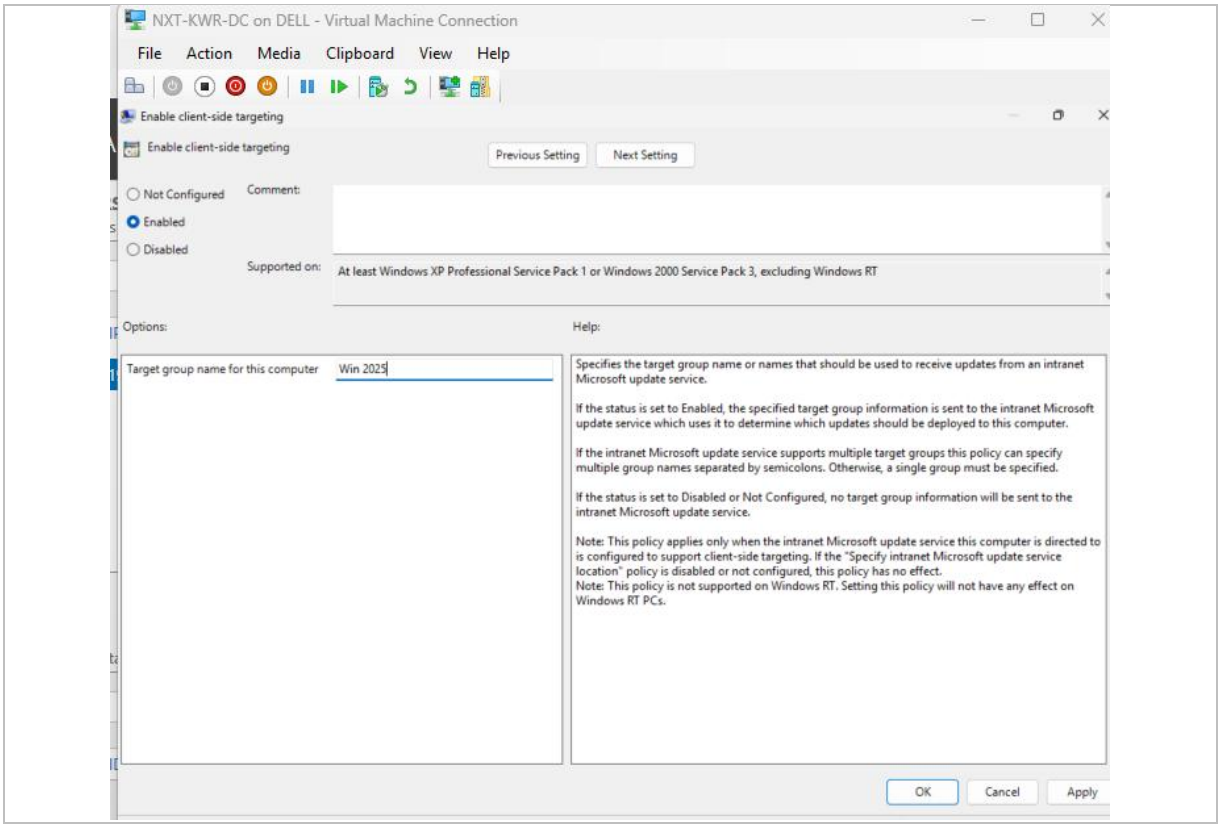


Figure 31: Enabling client-side targeting with a target group name

Other update policy settings let you control things like: who is allowed to install updates, when updates are installed, and whether updates can be paused.

Configuring the Automatic Update Policy

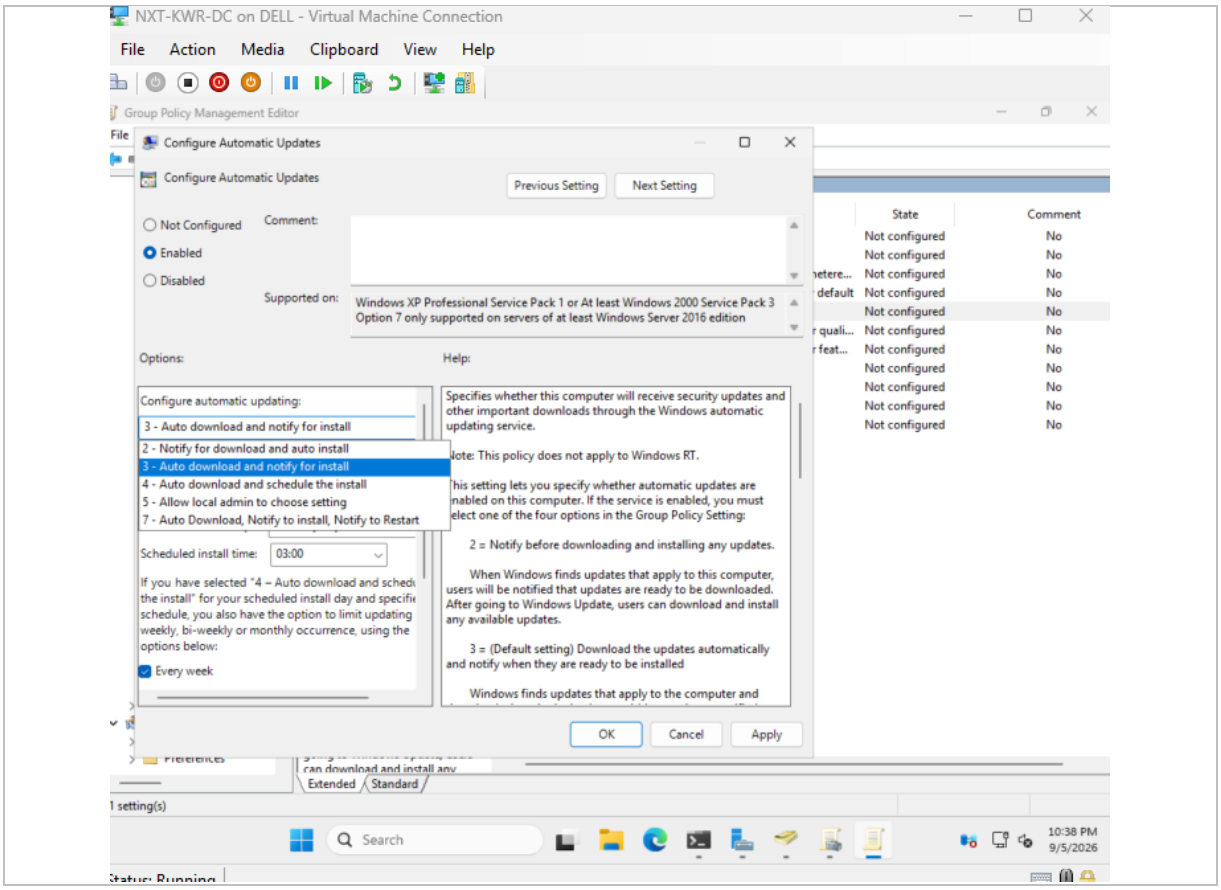


Figure 32: Configuring the Automatic Updates policy

When creating policies, it is best practice to assign different settings to client PCs and to servers, because their needs are different:

Computer type	Recommended setting
Client PCs	Usually set to auto-install updates and restart automatically once a week.
Servers	Usually never set to auto-install. Instead, use auto-download, then notify the admin before installing, and notify the admin before restarting. This avoids unexpected server reboots.

Table 3: Recommended Automatic Updates behavior

Once the policy is created, link it to the correct OU. You can also force a client to check policy immediately using the command line:

```
gpupdate /force
```

Computers that have not been assigned to any group appear under “Unassigned Computers” in the WSUS console. If you set up client-side targeting (explained above) and gave a group name, computers will instead appear in that named group.

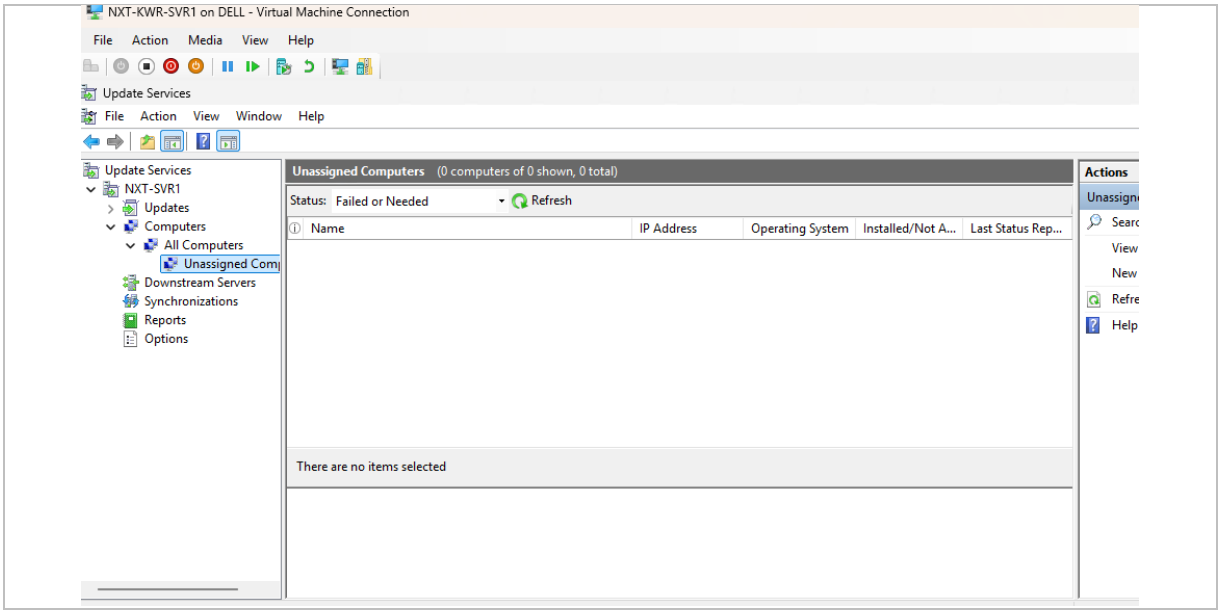


Figure 33: WSUS console — Unassigned Computers group

11. Approving and Assigning Patches

Patches can be approved manually or automatically.

Manual approval

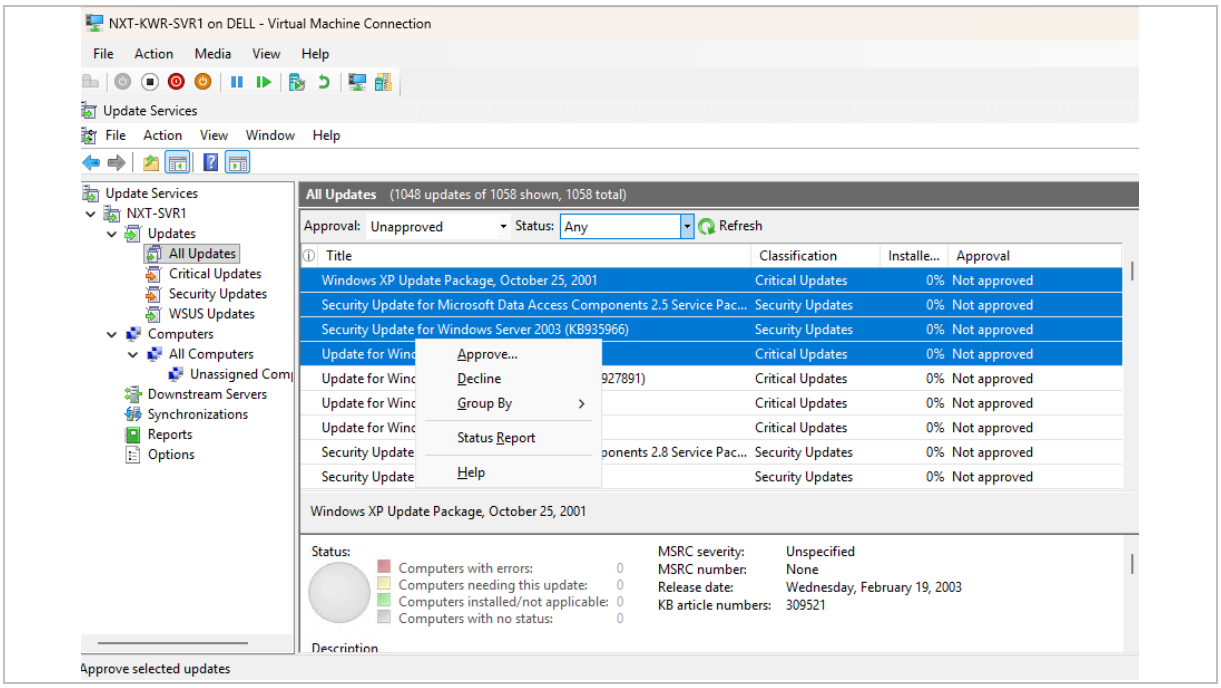


Figure 34: List of updates ready for manual approval

Select the patch you want, right-click it, and click Approve. The patch will then download, and you assign it to the target computer group.

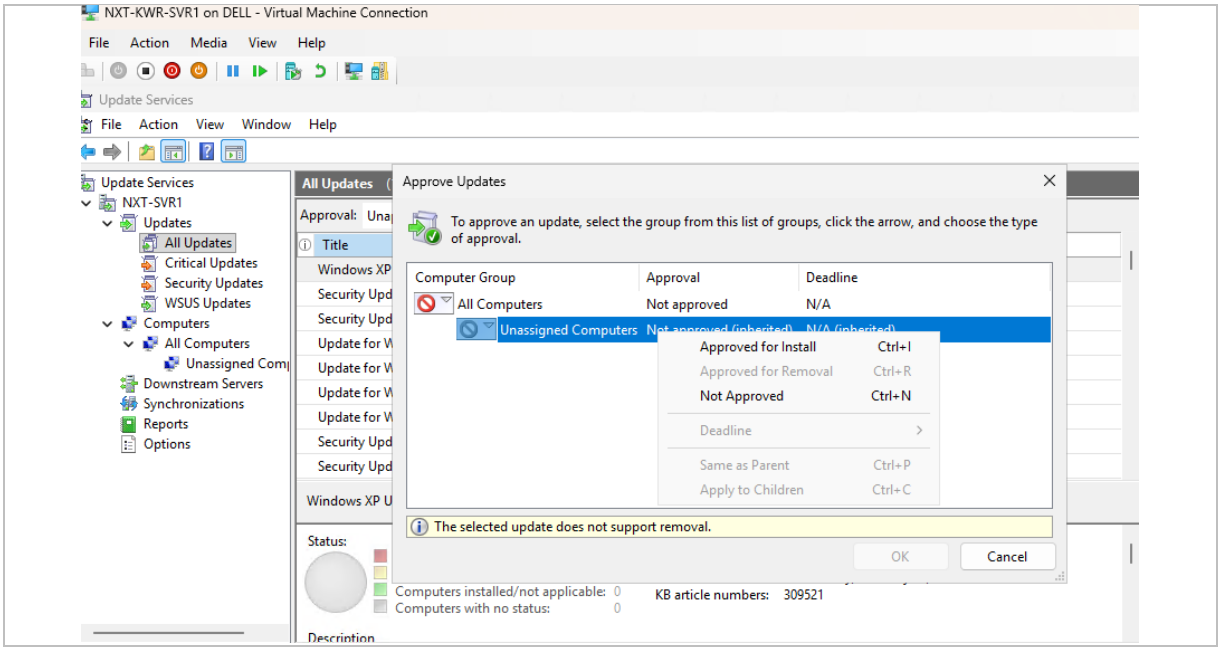


Figure 35: Approve Updates dialog — assigning an update to a computer group

Automatic approval

You can also set up rules so patches are approved automatically, without you doing it by hand each time.

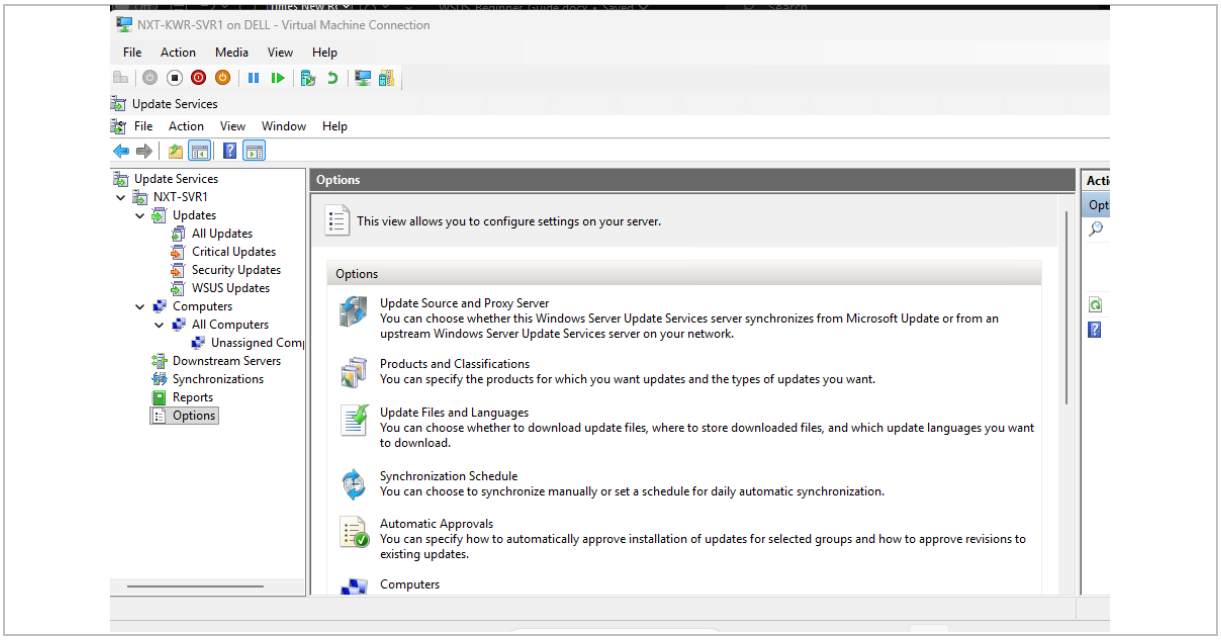


Figure 36: WSUS Options — Automatic Approvals

- Click Options, then click “Automatic Approvals”.
- Click “New Rule”.

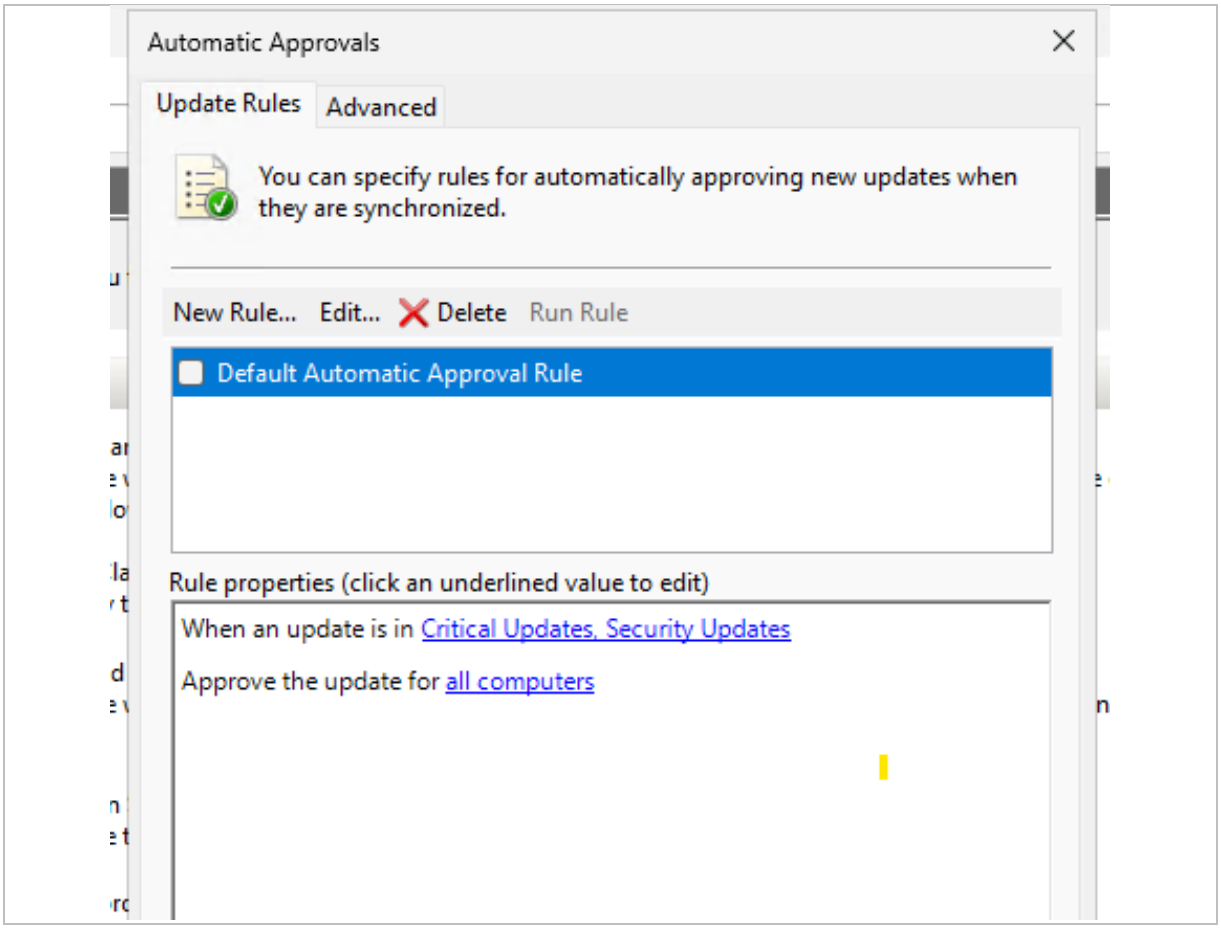


Figure 37: Default Automatic Approval Rule

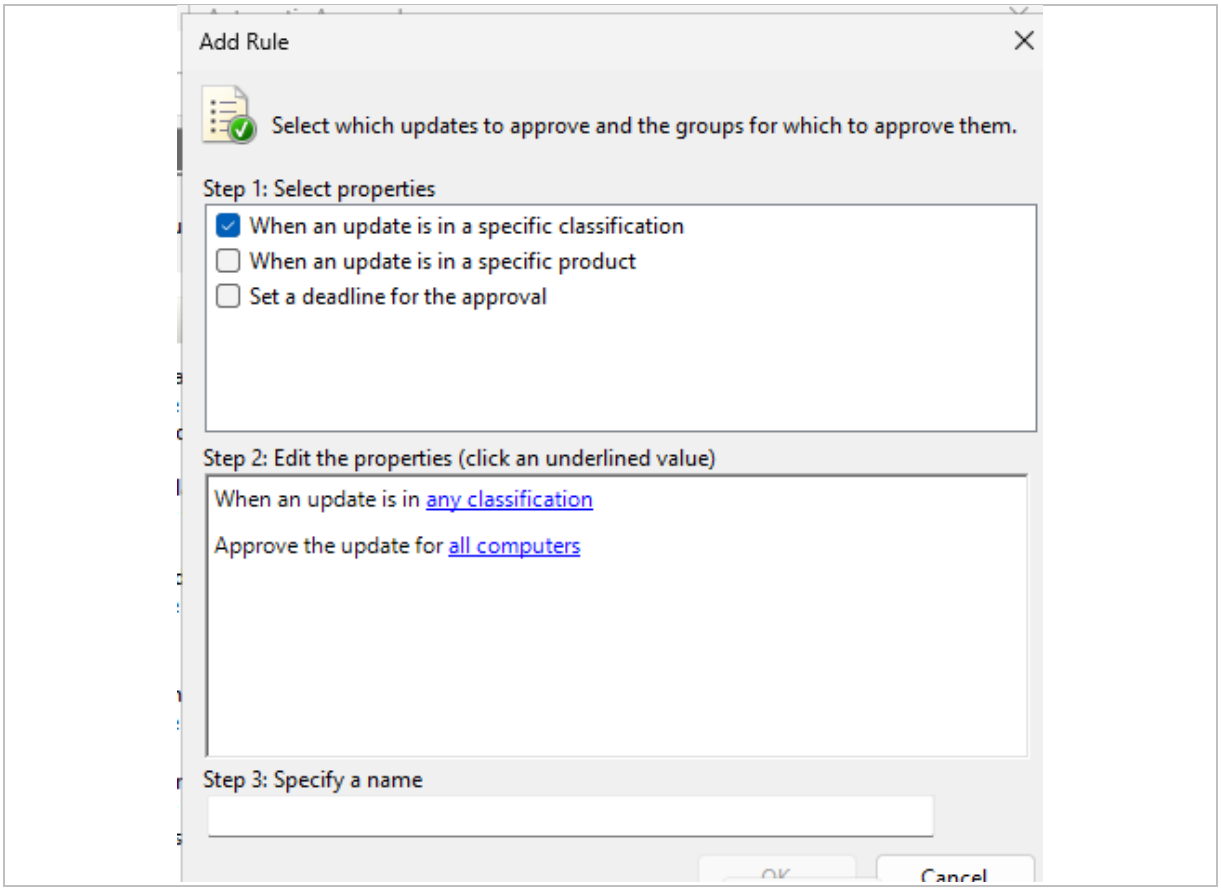


Figure 38: Add Rule dialog for a new automatic approval rule

This rule means: whenever a critical or security update is released for a specific product (for example, Windows Server 2019 or Windows Server 2022), it should be automatically approved for the matching computer group (in this example, the “Windows Server 2019” group).

> **NOTE** *When creating this rule, make sure to tick both: “When an update is in a specific classification” and “When an update is in a specific product”. Both conditions should be selected so the rule only applies to the updates you actually want.*

Summary: to manage clients in a corporate environment, you can use either Group Policy (as shown above) or a dedicated WSUS Agent/client software, depending on your company's setup.